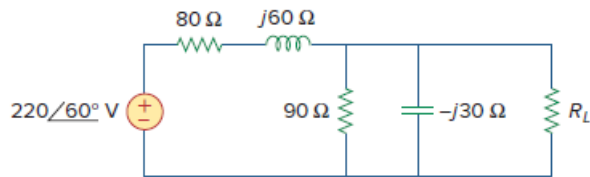


Problem

In Fig. 11.12, the resistor R_L is adjusted until it absorbs the maximum average power. Calculate R_L and the maximum average power absorbed by it.

Figure 11.12:

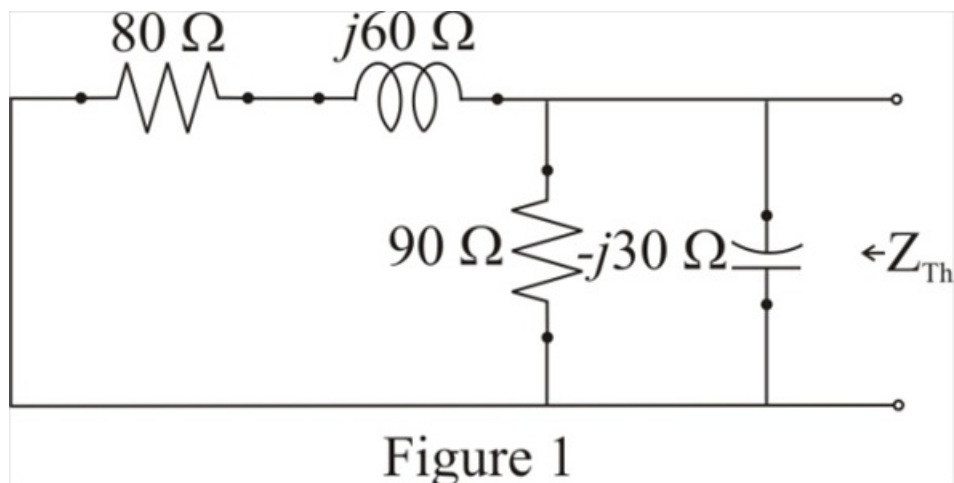


Step-by-step solution

Step 1 of 7

Refer to Figure 11.4 in the text book.

Deactivate the voltage source and disconnect the load resistance. Consider the circuit diagram shown in Figure 1.



Step 2 of 7

Calculate the value of the Thevenin's impedance Z_{Th} .

$$\begin{aligned}
 Z_{Th} &= (80 + j60) \parallel 90 \parallel (-j30) \\
 &= \frac{(80 + j60)(90)(-j30)}{(80 + j60)(90) + (-j30)(80 + j60) + (90)(-j30)} \\
 &= \frac{162000 - j216000}{7200 + j5400 + 1800 - j2400 - j2700} \\
 &= \frac{162000 - j216000}{9000 + j300}
 \end{aligned}$$

$$Z_{Th} = 17.18 - j24.57 \, \Omega$$

Therefore, the value of the Thevenin's impedance Z_{Th} is, $17.18 - j24.57 \, \Omega$.

Step 3 of 7

Consider the circuit diagram shown in Figure 2 to calculate the value of the Thevenin's voltage.

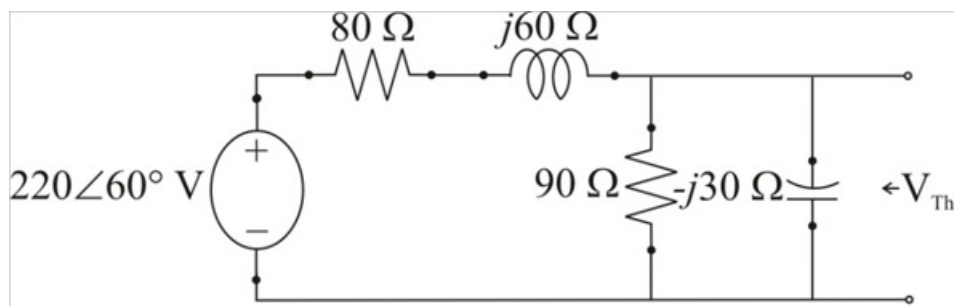


Figure 2

Step 4 of 7

Calculate the value of the Thevenin's voltage V_{Th} using voltage division rule.

$$\begin{aligned}
 V_{Th} &= (220\angle 60^\circ) \left(\frac{90 \parallel (-j30)}{80 + j60 + 90 \parallel (-j30)} \right) \\
 &= (220\angle 60^\circ) \left(\frac{\frac{(90)(-j30)}{90 - j30}}{80 + j60 + \frac{(90)(-j30)}{90 - j30}} \right) \\
 &= (220\angle 60^\circ) \left(\frac{\frac{-j2700}{90 - j30}}{80 + j60 + \frac{-j2700}{90 - j30}} \right) \\
 &= (220\angle 60^\circ) \left(\frac{9 - j27}{80 + j60 + 9 - j27} \right) \\
 V_{Th} &= (220\angle 60^\circ) \left(\frac{9 - j27}{89 + j33} \right) \\
 &= (220\angle 60^\circ)(-0.01 - j0.3) \\
 &= (220\angle 60^\circ)(0.3\angle -91.91^\circ) \\
 &= 66\angle -31.91^\circ \text{ V}
 \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, $66\angle -31.91^\circ \text{ V}$.

Step 5 of 7

Calculate the value of the load resistance R_L that will absorb the maximum power.

$$R_L = |Z_{Th}|$$

Substitute $17.18 - j24.57 \Omega$ for Z_{Th} .

$$\begin{aligned} R_L &= |17.18 - j24.57| \\ &= \sqrt{(17.18)^2 + (24.57)^2} \\ &= \sqrt{295.15 + 603.69} \\ &= \sqrt{898.84} \end{aligned}$$

$$R_L \approx 30 \Omega$$

Therefore, the value of the load resistance R_L that will absorb the maximum power is,

$$\boxed{30 \Omega}.$$

Draw the Thevenin's equivalent circuit diagram.

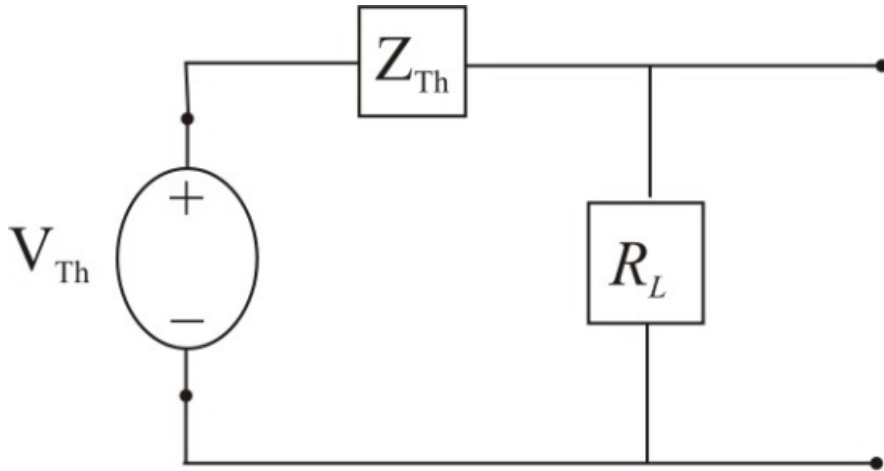


Figure 3

Step 6 of 7

Calculate the value of the current I flowing in the circuit.

$$I = \frac{V_{Th}}{Z_{Th} + R_L}$$

Substitute $66\angle -31.91^\circ \text{ V}$ for V_{Th} , $17.18 - j24.57 \Omega$ for Z_{Th} and 30Ω for R_L .

$$\begin{aligned} I &= \frac{66\angle -31.91^\circ}{17.18 - j24.57 + 30} \\ &= \frac{66\angle -31.91^\circ}{47.18 - j24.57} \\ &= \frac{66\angle -31.91^\circ}{53.19\angle -27.51^\circ} \\ &= 1.24\angle -4.4^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I flowing in the circuit is, $1.24\angle -4.4^\circ \text{ A}$.

Step 7 of 7

Calculate the value of the maximum average power $P_{\max}$ absorbed by the load resistance.

$$P_{\max} = \frac{1}{2} |I|^2 R_L$$

Substitute 1.24 A for $|I|$ and $30 \, \Omega$ for R_L .

$$\begin{aligned} P_{\max} &= \frac{1}{2} (1.24)^2 (30) \\ &= (1.5376)(15) \\ &= 23.06 \text{ W} \end{aligned}$$

Therefore, the value of the maximum average power $P_{\max}$ absorbed by the load resistance is,

$$\boxed{23.06 \text{ W}}.$$